

Name: M Abdullah

CMS:501223

Accuracy Assessment Lab Report

Steps Performed

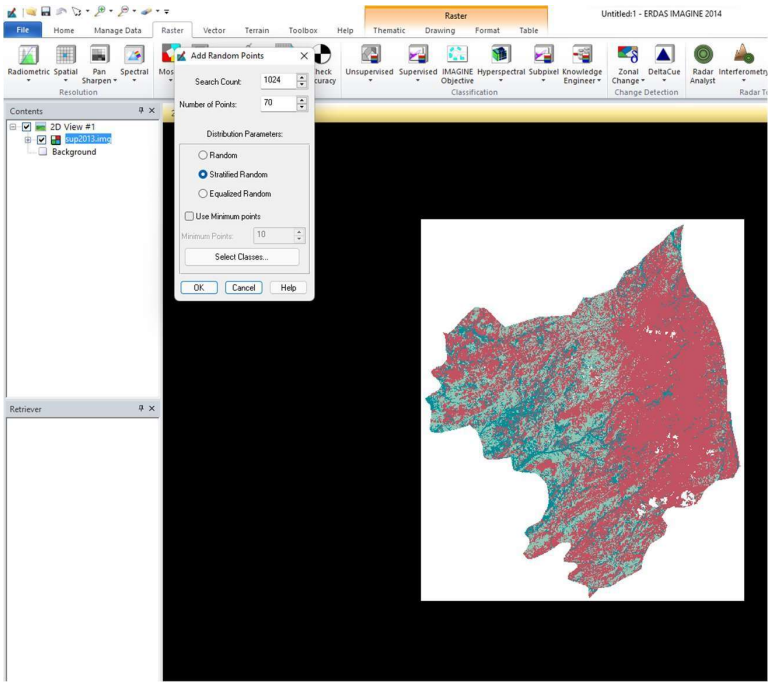
1. Loaded Classified Image
Opened the file *sup2013.img* in ERDAS Imagine and displayed it in a 2D viewer.

2. Opened Accuracy Assessment Tool
Selected **Raster → Supervised → Accuracy Assessment** to bring up the accuracy assessment dialog.

3. Linked the Viewer
Used **View → Select Viewer**, then clicked inside the classified image viewer to link both windows.

4. Opened Classified Image Inside the Tool
Used **File → Open** in the Accuracy Assessment window to load *sup2013.img* again.

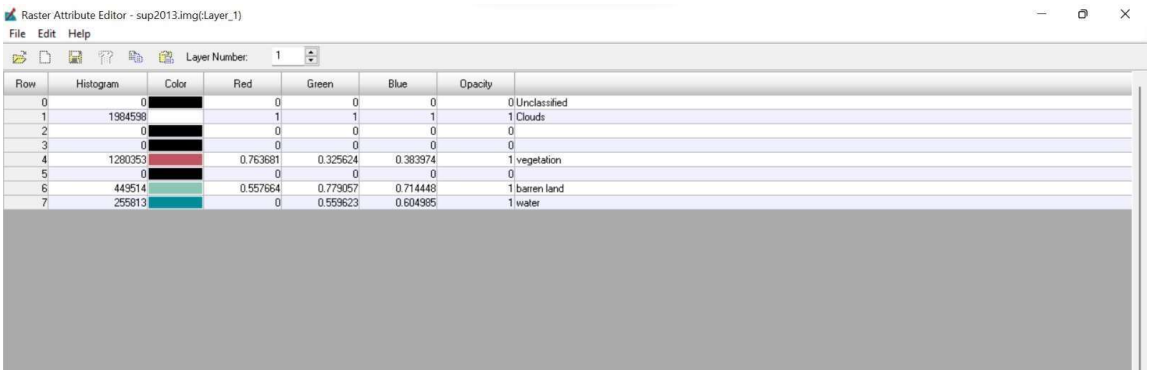
5. Created Random Sample Points
Selected **Edit → Create/Add Random Points**, chose **70 points**, **Stratified Random**.



6. Hid Classified Values
Used **Edit → Hide Class Values** to avoid bias when determining reference data.

8. Loaded Reference Image
Opened another reference viewer with *sup2013.img* stacked bands and linked both viewers.

9. Assigned Reference Data
For each sample point: zoomed to its location, interpreted the correct land-cover type, and entered the reference class value.



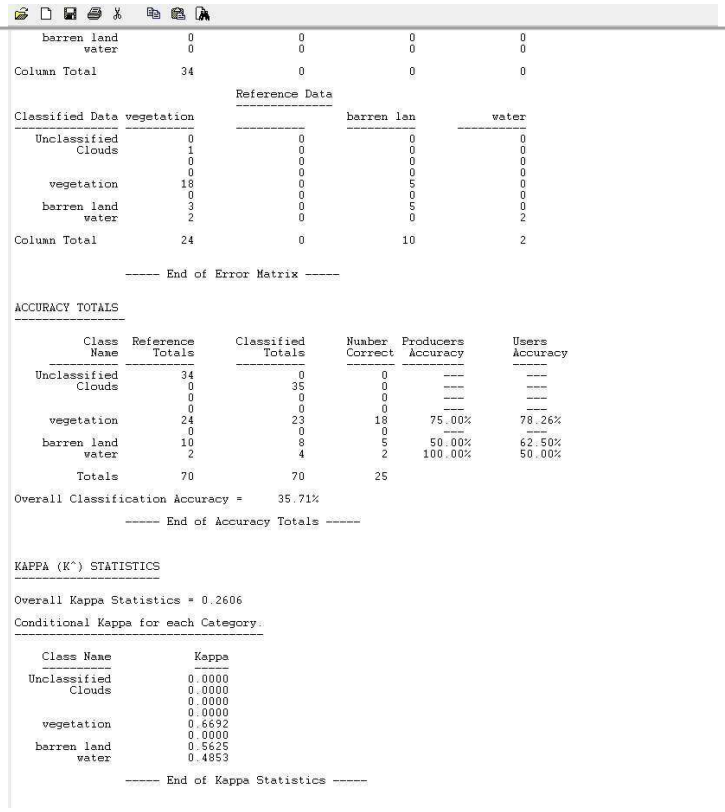
Row	Histogram	Color	Red	Green	Blue	Opacity	
0	0		0	0	0	0	Unclassified
1	1984598		1	1	1	1	Clouds
2	0		0	0	0	0	
3	0		0	0	0	0	
4	1280353		0.763881	0.325624	0.383974	1	vegetation
5	0		0	0	0	0	
6	449514		0.557664	0.779057	0.714448	1	barren land
7	255813		0	0.559623	0.604985	1	water

10. Generated Accuracy Report
Finally selected **Report → Accuracy Report** to produce the full error matrix, class accuracies, and Kappa statistics.

Summary of Output

Your error matrix included the following key results:

- **Overall Accuracy: 35.71%**
- **Classes with highest Producer's/User's Accuracy:**
  • *Water* → 100% Producer's accuracy
  • *Vegetation* → 75% Producer's accuracy
- **Lowest performing classes:**
  • *Unclassified & Clouds* → 0% (as expected—no reference points)
  • *Barren Land* → 50% accuracy
- **Overall Kappa Statistic: 0.2606**
- Conditional Kappas ranged from near 0 to about 0.66 (vegetation performed best).



barren land	0	0	0	0
water	0	0	0	0
Column Total	34	0	0	0

	Reference Data			
Classified Data	vegetation		barren lan	water
Unclassified	0	0	0	0
Clouds	1	0	0	0
	0	0	0	0
	0	0	0	0
vegetation	18	0	5	0
	0	0	0	0
barren land	3	0	5	0
water	2	0	0	2
Column Total	24	0	10	2

----- End of Error Matrix -----

ACCURACY TOTALS

Class Name	Reference Totals	Classified Totals	Number Correct	Producers Accuracy	Users Accuracy
Unclassified	34	0	0	---	---
Clouds	0	35	0	---	---
	0	0	0	---	---
	0	0	0	---	---
vegetation	24	23	18	75.00%	78.26%
	0	0	0	---	---
barren land	10	8	5	50.00%	62.50%
water	2	4	2	100.00%	50.00%
Totals	70	70	25		

Overall Classification Accuracy = 35.71%

----- End of Accuracy Totals -----

KAPPA (K^) STATISTICS

Overall Kappa Statistics = 0.2606

Conditional Kappa for each Category.

Class Name	Kappa
Unclassified	0.0000
Clouds	0.0000
	0.0000
	0.0000
vegetation	0.6692
	0.0000
barren land	0.5625
water	0.4853

----- End of Kappa Statistics -----

Accuracy Assessment Review Questions

1. Interpretation of the Error Matrix

- **Best Classified Class:**
  Vegetation shows the strongest performance overall, with 75% producer's accuracy and a relatively high user's accuracy. Water also has a perfect producer's accuracy (100%), though its user's accuracy is only 50%, meaning some non-water pixels were mislabeled as water.
- **Worst Classified Class:**
  Unclassified and *Clouds* technically have 0% accuracy, but the meaningful "worst" performing real class is *Barren Land* or the class with high confusion—only about 50% accuracy.
- **Confusions:**
  Several classes appear confused with one another, especially vegetation and barren land. Some points classified as one class show up as another in the reference data. This is common when spectral signatures overlap or the imagery has mixed pixels.

Overall, the matrix shows that the classification struggles with several land-cover types, leading to a relatively low overall accuracy of 35.71%.

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2. One Use I Would Be Comfortable With, and One I Would Not

Comfortable Use:
I would feel comfortable using this classified image for **general landscape composition**—for example, estimating roughly how much area is vegetation vs. barren land. Even if not perfectly accurate, the classification still provides a broad view of major land-cover categories.

Not Comfortable Using It For:
I would *not* rely on this map for **detailed planning or management decisions**, such as determining exact boundaries of land-cover types or tracking environmental changes over time. The relatively low accuracy and moderate class confusion mean it is unsuitable for precise mapping.

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3. What Is the Kappa Statistic and What Does It Tell You?

The **Kappa statistic** measures how much better the classification performs compared to **random chance**. A value of:

- **1.0** means perfect agreement
- **0.0** means the classification is no better than random guessing
- **Negative values** mean it is worse than random

Your Kappa value of **0.2606** suggests **low agreement**, but still better than chance. It confirms that the classification has meaningful structure, but there is considerable room for improvement.